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Augmenting Social Bot Detection with Crowd-Generated Labels

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Abstract. Social media platforms are facing increasing numbers of cyber-adversaries seeking to manipulate online discourse by using social bots (i.e., social media software robots) to help automate and scale their attacks. Likewise, some social media users can identify social bot activity at varying degrees of confidence. In this research, human reactions to social bot messages are used to augment existing social bot detection capabilities. Speech act theory is used to inspire a framework for assessing the credibility of instances where users identify potential bot activity, as not all user responses are of equal credibility for assisting with the bot detection task. The framework is then operationalized through deep learning methodologies to develop a computational system for identifying social bots. Real-world performance and practicality of the developed framework is demonstrated on a live, crowd-sourced data set collected from a real-world social media platform. Results show that consideration of crowd reactions to suspected bots can significantly improve bot detection performance. Furthermore, consideration of speech acts to evaluate crowd reactions can even further augment the system's performance, although speech acts themselves are not necessary to observe performance boost through crowd intelligence. This study serves as a grounding point for future work that can explore an augmented model for detecting other forms of algorithmically generated content within social media platforms.

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1. Introduction

The widespread reach and appeal of social media attracts numerous cyber-adversaries seeking to manipulate online discourse for their own benefit. Further exacerbating this issue, many cyber-adversaries are increasing their utilization of bots to help automate and scale their attacks (Ferrara et al. 2016). For example, an online public forum operated by the Federal Communications Commission (FCC) was recently attacked by a flood of bots that spammed anti–net neutrality comments, drowning out the voices of real citizens who supported opposing views (Lapowsky 2017). The proliferation of social bots (i.e., social media bots) is a gateway for potential exploitation of unsuspecting users who may interact with social bots thinking that they think are communicating with a human. Fortunately, although social bots do successfully deceive many, there is a growing number of users who are becoming increasingly aware of social bot activity. These users are sometimes able to identify and call out bots for instances of social media manipulation. Analysis of such human–bot interaction, however, is limited in existing literature. This research seeks to address this major knowledge gap by scrutinizing the efficacy of using real-world user reactions to suspected social bots to develop a feature set for augmenting traditional systems for social bot detection. Furthermore, one nontrivial issue that arises in this approach is that the capability for users to detect social bots is not equal, and thus they do so with varying levels of credibility. As a result, this study uses speech act theory as a foundation for building a framework to prioritize different types of human responses to bot activity for augmenting bot detection systems.

The proliferation of social bots presents severe ethical and practical challenges that warrant study. Much research thus far has focused on developing heuristics for detecting bots by scrutinizing their behaviors for highly synchronous, nonhuman activity patterns (Stieglitz et al. 2017a). For example, bots can be detected by scrutinizing user account behavior for use session lengths that are exceedingly lengthy and uncharacteristic of human activity. However, such advances in bot detection have not gone without response from adversaries that author social bots. Recent years have witnessed the emergence of many sophisticated, adversarial bots that possess evasion and antidetection capabilities (Ji et al. 2016). The ongoing arms race has prompted for many to call for more research on methods to inhibit the influence of social bots (i.e., social media bots) and their activities. For example, Twitter recently solicited research proposals for measuring online conversational health and combatting social media abuse, spam, and manipulation. Social bot research also frequently calls for engineering of new features to continue combatting against increasingly advanced bots (Yang et al. 2019).

Increased media exposure of social bots and their increased visible impact on the real world (e.g., FCC net neutrality comments) have made human users become more cognizant that social bots possess influence on online conversation. As a result, some social media users will now leave comments to alert other users of potential bot-like activity when suspicious contents are encountered. Although much social bot research already scrutinizes bot-generated contents, none consider incorporating the human-cognitive power offered within user messages associated with potential bot contents. Even among the latest studies that leverage advanced text mining, this crowd-sourced labeling has typically gone unused (Garcia-Silva et al. 2019).

Extending current social bot detection strategies to include such crowd-sourced labels is a nontrivial effort. For example, consider a scenario where two human users claim they have identified social bot activity. One user may proclaim a statement such as, "I have definitely found a bot," whereas the other questions "I think I may have found a bot?" Although both these data points are human-curated, it could be argued that one data point provides a higher level of confidence or credibility than the other. Likewise, their influence on the bot detection task should then be weighed differently. This raises the question of how to determine the influence of each user reply in a logical, systematic way that avoids ad hoc reasoning.

In this research, speech act theory is used as a vehicle to measure and weigh the responses of different users. Each user comment can be assessed through determination of what action the comment is trying to accomplish, for example, when a user declares that they have identified a bot account or when asking other users to share their judgments on whether a suspicious account is indeed a social bot. This perspective for incorporating user replies in social bot detection yields superior performance over traditional models.

The major contribution of this work is to pursue a strategy of social bot detection that involves evaluation of crowd reactions. This is a gap in literature as many extant bot detection systems make use of benchmark data sets that contain messages only produced by known bot accounts and do not include replies or crowd reactions to bot-generated content (e.g., the data set provided by Cresci et al. 2017). Conversational replies between social bots and real users are used for bot detection within this work, presenting a novel strategy for bot detection by considering the reactions of other users. Furthermore, this study goes one step further by not only incorporating crowd reactions for bot detection, but also demonstrating a strategy by which crowd reactions can be evaluated for their certainty. This is operationalized by use of speech act theory. Although speech act theory has been useful in many information systems (IS) research contexts, it has not previously been used for evaluating crowd-sourced labels or within the greater body of security informatics research. Whereas speech act theory has been primarily used in research to develop sense-making capabilities for conversational transcripts or other text-based conversations (Abbasi et al. 2018), its use in this work to act as a mechanism of weighting the credibility of user labels presents a novel application. An outcome of this work is evidence that future bot detection research should consider incorporating crowd reactions as a feature set to identify potential bots and that speech acts may be a mechanism for evaluating the certainty of crowd reactions.

Our work lies at the intersection of combining human- and machine-cognitive power. The system design instantiated in this work uses state-of-the-art machine learning techniques and attempts to solve a problem of growing interest to both academia and industry. The need for more computational work of this nature is echoed in many recent calls for more design science work in premier IS journals (Agarwal and Dhar 2014, Goes 2014, Rai 2017). Additionally, this research lends itself to investigations of how real users identify social bots during their interactions on social media and how such interactions can be leveraged to more rapidly identify malicious bots.

The rest of this paper is organized as follows: first, relevant background is summarized to provide context to this study's problem domain and methodological focus. Next, a research framework using an augmented intelligence approach is introduced. This framework incorporates many of the latest techniques in computational linguistics and makes use of speech act theory as a well-established grounding when assessing human replies to bot-generated messages. After the framework is elaborated, detailed descriptions of the data set, experiment design, and evaluation are provided. Then follows a discussion regarding the key conclusions and contributions that result from this work. This study features many practical contributions to the social bot research stream and research that uses crowd-generated labels and ground truth.

2. Background

To help guide the development of this study, three relevant streams of literature are reviewed. First, we contextualize this study with the latest developments in social bot research to highlight the current state of detection capabilities while also making visible current research gaps of interest. Second, recent studies focused on social media analytics are reviewed to help inform system design within this study. Finally, a review of speech act theory is provided as it is a promising instrument for the context of this study and has seen use in previous IS and social media analytics literature (Ludwig and de Ruyter 2016, Abbasi et al. 2018).

2.1. Social Bots

2.1.1. Overview. Social bots have emerged as a major topic of focus in recent years because of their increasingly noticeable impact on daily life. Recent estimates have placed social bots to comprise 5%–8% of total users in some of the most popular social networking platforms (de Lima Salge and Berente 2017). Many of these bots behave maliciously by disseminating spam, spreading false information and propaganda, limiting free speech, propagating malware, and as a delivery mechanism for phishing attempts and other scams (Oentaryo et al. 2016, Efthimion et al. 2018). Because of their increasing prevalence and often adversarial nature, social bots have evolved into a serious threat (Ferrara et al. 2016).

Social bots impact both online platforms and their users by influencing how and what information diffuses within their social network (Davis et al. 2016, Subrahmanian et al. 2016, Varol et al. 2017, Yang et al. 2019). Social bots may also take advantage of social networking features to build stronger ties with other users and extend their reach. Thus, not only will users be burdened with sifting through publicly posted bot-generated messages to find real content, they may also face increases of spam and illegitimate social networking requests directly from social bots themselves. If the growth of social bot activity is left unchecked, users suffer as the quality of the platforms they use become debased with low-quality and false content (Yang et al. 2019, Pozzana and Ferrara 2020). The overall user experience becomes less meaningful as it becomes more difficult to identify and consume quality content. The reduced content utility and user experience lead to user attrition and losses for the online platform.

The menace of social bots has come at a time when social media and mainstream Internet communities are seeing increasing amounts of "fake news" or misinformation, spam, and scams. Social bots echo many of the same characteristics of these issues while also maintaining enough nuances to present a distinct set of adversarial activity on social media. As such, social bot detection studies can borrow from related research streams but still require their own logical, systematic approaches to solve context-specific problems.

Problems such as fake news detection, spammer detection, and so on, are typically addressed through scalable computational frameworks that use advanced text analyses to detect adversarial behaviors within real-world environments (Ruchansky et al. 2017, Jain et al. 2019). The pipeline to collect, process, and analyze data would be shared across these contexts, even extending to the feature sets and methodologies used to represent and model data. There are shared characteristics of the problem domains that can help justify assumptions, proposed approaches, and system design choices.

2.1.2. Types of Social Bots. Social bots can appear in many forms depending on the goals and intentions of their authors, leading to a variety of bot behaviors and actions (Subrahmanian et al. 2016, Ferrara 2018). Most social bots are intended to spam messages that propagate some political or social position. For example, some bots will mimic human behavior to manufacture seemingly legitimate support for various perspectives such as grassroots political campaigns, extremist propaganda, unfounded rumors and conspiracy theories, stock market manipulation, and more (Varol et al. 2017, Ferrara 2018).

Social bots are also able to form many social network ties with other users through social functions such as "following" other users and interacting with their postings. These behaviors increase the volume and reach of social bots' messaging, leading to adverse outcomes for both the social platform and its users (Ji et al. 2016, Stieglitz et al. 2017b, Shi et al. 2019). Less advanced bots will create posts at predictable intervals with little variation in postings, whereas more advanced bots mimic human behavior. Detection of advanced bots that successfully mimic human social media patterns is a nontrivial problem and the focus of many researchers (e.g., the Defense Advanced Research Projects Agency (DARPA) challenge highlighted in Subrahmanian et al. 2016).

2.1.3. Social Bot Research Methods. Prior studies that focus on the social bot problem often use similar computational research methods and system design. The primary points of overlap include both assumptions and techniques regarding ground truth (i.e., whether a datapoint is really bot-generated or not), how feature sets are developed, which modeling techniques are used, and what evaluation procedures are performed.

Establishing ground truth of which records are algorithmically generated, as opposed to being created by real human users, is a common and nontrivial challenge to solve in the bot detection research stream (Davis et al. 2016, Subrahmanian et al. 2016, Varol et al. 2017). There exist few large, open data sets with complete, verified ground truth in this context (Cresci et al. 2017). Many studies recognize that ground truth is extremely difficult to obtain in this research context but that there is still merit in trying to identify some potential bots so that it is possible to research and construct bot detection tools (Yang et al. 2019). The methods used to identify potential bots are numerous. Some studies deploy "social honeypots" or social media accounts that are created to passively sit and await interaction (e.g., messages, "likes," and "follow" requests) from suspected bots (Subrahmanian et al. 2016). Other researchers make use of public datasets that they assume as ground truth for the purposes of enabling the practical research efforts (Davis et al. 2016, Varol et al. 2017). Crowd-sourced data sets are also an option (Ferrara 2018). Finally, some pursue the option of hiring expert annotators to manually sift through collected social media data and use human judgements to label datapoints as bot- or human-generated (Ferrara et al. 2016). All three techniques have their inherent biases and limitations, and no technique is arguably better than the other. The "right" data set is dependent on the objective of the study. For example, using honeypots allows for greater volumes of collected social bot activity; however, bots identified through honeypots may be exhibiting obvious amounts of spam activity. Conversely, human annotators may be able to identify some less obvious bots that exhibit more human-like characteristics, but manual vetting can be slow and is not scalable.

Regarding analytical methodology, feature generation and model selection are closely intertwined in the bot detection context. For example, content-based detection methods may make use of content analysis methodologies (e.g., text mining), whereas network-based detection would rely on scrutinizing network behaviors and related metrics. Detecting bots through content analysis often includes scrutinizing content for repetitive messages, frequent language use changes, high volumes of messages and sentiment regarding particular topics (e.g., politics, advertisements, scams, extremist perspectives, etc.), and more (Ferrara et al. 2016, Varol et al. 2017). Network analysis features may include centrality measures, follower (i.e., social tie) networks within online communities and analysis of follower/followee metadata (Yang et al. 2019). The extracted features are then used for training a machine learning classifier to distinguish between humans and bots. It should be noted that the social platforms play a major role in what features are available for social bot detection; for example, Twitter supports "retweet" functionality that is a commonly observed feature, but there is no analog or equivalent feature supported on platforms such as YouTube or Reddit. Furthermore, many platforms have their own unique characteristics and interfaces that may influence user discourse in ways that cannot be easily generalized to other platforms. For example, there are character limits imposed on Twitter messages that undoubtedly impact how users converse on the platform, but no such character limits exist on Reddit, thus allowing users to craft messages more freely. Overall, this study focuses on Reddit to scrutinize bot detection within a platform that supports lengthy online conversations over punctuated discourse. Thus, strong methodological focus is placed on text mining, and a review of related works is necessary.

Table 1 contains a summary highlighting some representative works of recent, high-performance techniques for text-based social bot detection. The highlighted studies help showcase what techniques and performance recent works have achieved, given varying contexts to operationalize bot detection systems. A strong majority of social bot detection studies do focus on Twitter, thus biasing developed models for short-form social media; it is unclear what the efficacy of these previous efforts would be if applied to longer-form social media, such as on Reddit. Surprisingly, application of social bot detection methods to long-form social media appears to be lacking. Some studies do indeed use Reddit data but within the context of scrutinizing bot-account activity patterns or network ties rather than any text content analysis (Ferraz Costa et al. 2015, Hurtado et al. 2019).

The studies cited in Table 1 pursue a similar pipeline for constructing social bot detection systems. First, a data set is manually curated to develop ground truth. The featured studies either label their own data or borrow from publicly available data sets (Cresci et al. 2017, Varol et al. 2017). Next, many studies adopt text analyses to some degree, some relying on more basic techniques such as lexicons to analyze text, whereas others use state-of-the-art models such as Recurrent Neural Networks (RNN) or transformers. In more recent works, pretrained language models are often used rather than learning embeddings and models from scratch because of their demonstrated superior performance (Garcia-Silva et al. 2019). Beyond text, some studies consider other feature categories that span network characteristics, account-level behaviors, and more (Cai et al. 2017, Varol et al. 2017, Heidari and Jones 2020). However, as mentioned previously, many such features adopted in prior studies are often platform specific and not generalizable (e.g., retweets). There is a clear lack of recent studies that can be generalized away from Twitter/microblogs and applied to other social platform contexts. This gap could lead to potential model development that fails to scale successfully across social platforms possessing characteristics quite different from microblogs, creating a need for bot detection efforts that are focused on different types of platforms.

Another limitation is the very fact that many studies make use of the data set available in Cresci et al. (2017) to develop their research. The Cresci et al. data set possesses some gaps that inherently limit the potential contributions of future studies. In particular, this data set provides message-level granularity rather

Table 1. Recent Works on Social Bot Detection Using Text Analytics

Paper	Account- or message-level classification?	Method(s) used	Feature(s) used	Data set/platform	Performance achieved (F-measure)
Heidari and Jones 2020	Message	GloVe and ELMO both utilized to develop tweet embeddings; various models for binary classification of bots	Tweet message content and sentiment	Twitter (2,839,361 human messages; 3,457,344 bot messages)	0.976
Garcia-Silva et al. 2019	Account	A survey of pretrained models including some RNNs and Transformers (e.g., BERT)	Tweet message content	Twitter (1,208 humans posting 279,495; 722 bots posting 220,505 tweets)	0.8388 with base BERT; 0.8074 with LSTM)
Wei and Nguyen 2019	Account	Pretrained Word Embeddings (GloVe) + BiLSTM	Tweet message content	Twitter (3,474 human accounts, 1,455 bot accounts)	0.9262 for account-level detection
Kudugunta and Ferrara 2018	Both	Pretrained Word Embeddings (GloVe) + BiLSTM	Tweet message content	Twitter (3,474 human accounts posting 8,377,522 tweets, 4,912 bot accounts posting 3,457,344 tweets)	0.9900 for account-level detection; 0.9600 for message-level detection
Cai et al. 2017	Account	Word Embedding s + LSTM	Tweet message content and temporal behaviors	Twitter (2,742 human accounts and 2,916 bot accounts)	0.873 for account-level detection
Varol et al. 2017	Account	Random Forest	Account behavior features, network features, and some text analysis	Twitter (3,000 accounts at varying probabilities of being bots, 200–300 messages per account)	Do not report F-measure, but report 0.95 AUC
Davis et al. 2016	Account	Random Forest	Account behavior, network, temporal features, basic NLP	Twitter (15,000 bot accounts; 16,000 human accounts)	Do not report F-measure, but report 0.95 AUC

than at the conversational or thread level, thus limiting possibilities to reconstruct conversations and scrutinize reply-to relationships. The issue in the original data set persists in multiple studies recently published in this research stream and demonstrates a significant gap in social bot detection capability. Utilization of crowd reactions could be particularly important because recent social bots generally feature sophisticated antidetection and evasion procedures that see continuous improvement (Ji et al. 2016, Stieglitz et al. 2017b). Many possess capabilities to receive instructions in real time from their owners, resulting in dynamic strategies and behaviors being deployed at the will of cyber-adversaries (He et al. 2017). Existing efforts have been unable to address such advancement and proliferation of social bots.

New perspectives for bot detection system design are necessary. As evidenced by Table 1, existing bot detection research is limited in that many studies only evaluate messages posted by suspected bot accounts and do not exploit crowd reactions containing human judgement. As real users become more aware of social bot activity, they may react to identified bots by posting messages that contain useful features for social bot detection. The application of crowd-generated labels for bot detection problems is novel and does not appear in prior literature to the best of the authors' knowledge. However, the use of crowd-generated labels in other domains has indeed occurred several times in the past (Kajino et al. 2012, Nguyen-Dinh et al. 2013). For example, crowdsourcing services such as Amazon Mechanical Turk have allowed researchers to use crowd intelligence for a variety of public health systems (Lasecki et al. 2013, Maier-Hein et al. 2016, Abad et al. 2022). Some other examples of recent scholarly works using crowd-generated labels have included identifying skillsets required for tasks (Leano et al. 2016) and labeling machine learning data sets (Chang et al. 2017). Across these studies, the labels were either explicitly asked for as part of a task assigned to users, or they were otherwise implicitly generated by simply observing the natural use of a system (Zhu et al. 2016). This research adopts a similar perspective; it is presumed that individuals' usage of social media may lead to similar implicit signaling of bot activity and that the efficacy of crowd-generated labels to augment social bot detection can thus be investigated. This reasoning leads to a focal research question.

Research Question 1. *What impact would analyzing crowd reactions have on the bot detection task?*

Deeper scrutiny of conversational-level features for bot detection appears to be still novel. A combination of analyzing long-form social media in conjunction with capturing conversational-level data to scrutinize reply-to relationships appears to be completely absent from extant literature. Scrutinizing conversational-level data can be key to developing next-generation social bot detection systems, as it becomes possible to observe natural user actions and behaviors toward suspected bots for further consideration. Additionally, conversational context enables a greater understanding of social bots' impact on users because direct interactions can be observed. These reasons necessitate the development of bot detection studies using varied data sets and targeting different platforms.

Use of more advanced computational methods and incorporating human messages associated with bot-generated content can yield many new opportunities to counter adversarial capabilities. In particular, other social media analytics research outside of the bot context has already adapted many of the latest advancements in statistical Natural Language Processing (NLP), benefiting from newer methodologies that are useful for the type of scalable text analyses necessary for this research. Other social media analytics work may also provide insights for operationalizing a framework that augments the traditional machine learning approach to social bot detection with human cognitive power.

2.2. Social Media Analytics

Social media is a frequent data source used in academic research. There exist several studies using social media data to investigate issues related to e-commerce, marketing, health, security, geopolitics, and more (Agarwal and Dhar 2014, He at al. 2017, Benjamin et al. 2019, Liu et al. 2020). In recent years, it has become more common for social media analytics research to use computational techniques for analysis of large-scale data sets. When reviewing recent works, two common applications for social media analytics emerge. First, many studies use a form of topic analysis to try and develop understanding of what social media participants converse about (Dutta et al. 2018, Liu et al. 2020). Topic analysis allows for one to quantify the topical content of a text document, enabling comparisons between documents, categorization, and more (Vaast et al. 2017, Abbasi et al. 2018). The second approach is sentiment analysis that allows for the quantification of the emotional content contained within a document (Dong et al. 2018). In social media analytics contexts, sentiment analysis is often paired with some form of topic analysis to identify user sentiment regarding a particular topic.

The popularity of topic and sentiment analysis can be largely attributed to their pragmatic usefulness and their ability to capture some of the most important aspects of language, semantics, and syntax (Winograd and Flores 1986). However, the combination of topic or sentiment analysis alone is not enough to fully operationalize a framework that augments traditional machine learning approaches for social bot detection with human intelligence by considering crowd reactions. Primarily, incorporating user certainty when identifying social bots is a nontrivial issue, as users will vary in their ability to accurately identify bot-like activity. Thus, a second research question of this study is as follows.

Research Question 2. *How can crowd reactions toward potential bots be evaluated in terms of certainty?*

A framework that successfully combines human and machine cognitive power by using crowd-sourced data must weigh all data points of interest, consider their credibility (as not all datapoints are equal), and learn from them accordingly. To the best of the authors' knowledge, this issue has not been explored previously in literature. Linguistic theory may be used to help computationally operationalize this task.

2.3. Speech Act Theory

2.3.1. Background. If topic and sentiment analysis can be thought of as tools to extract what people are saying, speech acts can be considered as a perspective that enables assessment of what action the speaker is intending to undertake (de Moor and Aakhus 2006). Speech acts are useful for identifying the underlying intent of any communication from one individual to another (Austin 1962, Searle 1969, Abbasi et al. 2018). Although many linguistic theories exist, speech acts are unique in proposing a vehicle for connecting language use to specific objectives that individuals wish to achieve when speaking. This perspective is useful for providing sense-making in social interactions between individuals (Lyytinen 1985, Kuo and Yin 2011).

There exist multiple speech act taxonomies, of which a few have risen to prominence (Moldovan et al.

2011). The most classic categorization of speech acts was introduced by Austin (1962) and posits five major speech act classes that categorically define different forms of verb usage. The five speech acts were *expositives* (verbs for asserting views) *exercitives* (verbs issuing a decision), *verdictives* (verbs for delivering a finding), *commissives* (verbs committing the speaker to some future action), and *behabitives* (verbs involving the reaction of the speaker to someone else's conduct).

Similarly, in the formalization of speech act theory introduced by Searle (1969), which is one of the more frequently cited perspectives, five speech categories are defined; they are called the assertive, commissive, declarative, directive, and expressive speech acts. In this taxonomy, the assertive speech act describes when an individual makes a statement that can be evaluated as true or false. The commissive speech act will commit the speaker to some future action. The declarative speech act is used by individuals generally stating some immediate change in affairs. The directive speech act is intended to have the speaker attempt to get some other individual to carry out an action. Finally, the expressive speech act is used when a person states some information regarding their emotional or psychological state. These five speech acts are elaborated further with examples in Table 2. Searle (1969) has been the basis of many computational works using speech act categorization (Qadir and Riloff 2011, Abbasi et al. 2018).

Another speech act categorization scheme was proposed by D'Andrade and Wish (1985) with a focus on closely integrating speech acts with multiple dimensions of interpersonal behavior. This formalization was also explicitly described as to only apply to native speakers of a language, which can be a constraining assumption for online discourse. The speech act categorizations in this formalization are assertions (expositives), questions (interrogatives), requests and directive (exercitives), reactions and expressive evaluations (behabitives), commitments (commissives), and declarations (verdictives and/or operatives).

Sometimes speech act formalizations can be made context specific to help describe the nuances of a particular setting. For example, a tutoring environment may require a different set of speech act categorizers to comprehensively describe the possible range of social interactions (Olney et al. 2003). In this case, a speech act taxonomy that divides questions into 16 subcategories while only maintaining three categorizations for nonquestions was developed (Olney et al. 2003).

Although most speech act formalizations are handcrafted after intensive literature review, philosophical reasoning, and study, there are other speech act formalizations that incorporate computational approaches. Computational approaches typically develop a formalization of speech acts that augments a human-built categorization scheme with machine-learned rules (Alexandersson et al. 1997). The objective of such efforts is to assist with building complex language modeling tools for dialogue processing or other tasks.

The *rational speech-act* (RSA) theory of language understanding was notable given its formalization of the perspective that humans optimally choose their speech to maximize a given objective (Goodman and Stuhlmüller 2013). RSA also argues that the listeners of a message acknowledge this notion and thus interpret another's speech by assessing both the semantics of their message and the social context to identify the speaker's purpose. This idea of extralinguistic knowledge (e.g., social context) being key to language understanding is one aspect that RSA attempts to formalize through Bayesian statistics and information theory (Goodman and Stuhlmüller 2013, Goodman and Frank 2016).

Many of the speech act formalizations overlap in inputs, methods, and applications. Syntactic and semantic information must typically be extracted from a language utterance to serve as inputs for analysis (Moldovan et al. 2011, Qadir and Riloff 2011). This can include extracting bigrams, parts of speech, punctuation, predefined keywords, and more. These datapoints serve as features to train a machine learning classifier that can categorize language into speech acts (Qadir and Riloff 2011, Abbasi et al. 2018). Such classifiers can be deployed in any scenario where computational analysis of text is

Table 2. Five Speech Acts of Speech Act Theory as Described in Searle (1969)

Speech act	Description	Examples	Example sentence
Assertive	A person states fact about the world.	Asserting, stating, describing	"Social bots are becoming more common"
Commissive	A person commits to some future action.	Pledging, threatening, offering	"I am going to report this bot"
Declarative	A person states a change of affairs effective immediately	Policies, pronouncements, verdicts	"This is definitely a bot account"
Directive	A person attempts to get others to carry out an action	Commands, suggestions, questions	"Can someone verify whether this is a bot?"
Expressive	A person expresses something regarding their psychological state	Apologies, complaints, thanking	"I feel there are bots everywhere now"

relevant. However, not all speech act taxonomies align to this workflow. For example, one of the assumptions of the RSA theory is that categorization of speech act is dependent on the intuition and perception of the recipient listening to the message in question (Goodman and Stuhlmüller 2013). Thus, in this model, the speech act arises from the listener's interpretation rather than the speech act being defined on utterance by its speaker. Although formalizations such as RSA theory have numerous applications such as in game theory, they are difficult to scale across networks of individuals that simultaneously communicate with one another while possessing little to no prior knowledge of each other, such as is the case with social media platforms.

2.3.2. Speech Acts for Bot Detection. Speech act categorization has not been explored for use in social bot detection. However, there is a precedence of social bot studies using automated text analyses for identification of potential bot activity. The analytical pipeline used for these prior text analyses can be potentially extended to incorporate speech act categorization. Data identification, collection, and processing would be similar as previously described with social bot research. However, data annotation and feature generation would be engineered to operationalize speech act categorization, resulting in a multiclass classification problem where each class corresponds to a distinct speech act.

As is the case with other social bot research, developing ground truth presents a nontrivial issue. One advantage to using crowd-sourced data sets as ground truth is the ability to scale the construction of data sets over time, with adequate volume. These crowd-labeled bots can be used to train classifiers and build systems with capabilities to identify other social bots, resulting in a successful coupling of human and machine cognitive power. However, users do not always report social bots they encounter because they may be uncertain and lack confidence in their judgements of whether an account is indeed a bot or not. Thus, not all crowd reactions are equal as they often differ on their credibility. Some crowd-sourced data will be more informative and of higher certainty. Selectively choosing the highest-quality crowd-sourced data as inputs for training a bot detection system may yield greater performance.

One potential solution to this issue is the use of speech acts for evaluating crowd reactions. Consider two traditional text mining approaches: topic and sentiment analysis. These approaches could only tell you whether a social media message discusses the topic of social bots and whether the message has positive or negative sentiment. These analyses do not provide much information regarding the underlying action of the message author and whether their assessment is credible. In such cases, speech act classification can help capture users' intent and provide additional context to train bot detection systems. This leads to the third research question of this study.

Research Question 3. *Can speech acts be used to computationally assess crowd certainty when identifying potential bots?*

In this research, human user replies to social bot content can serve as a form of crowd-labeling. The comments should be analyzed and weighed individually according to some metric of usefulness or credibility. Speech acts can be used as a vehicle to capture a level of intent and confidence in users' identification of social bot activity. User comments can be categorized by their speech act, providing yet another feature for bot detection. Experiments can be performed to evaluate the impact of different speech acts and which user comments exhibit the greatest qualities for improving detection performance. The use of speech acts for evaluating crowd labeled and crowd-generated ground truth is a novel application featured in this study. Speech act categorization can help identify the intent of user-generated content, and when paired with topic and sentiment analyses, would potentially allow for new use cases or enhanced performance in social bot detection and beyond. Any context where crowd-sourced messages are assumed to be of unequal quality may potentially benefit from speech act categorization.

3. Research Approach

The primary focus of this work is to explore computational solutions that may tip the balance of power between bot authors and platform owners. Currently there is an arms race between platforms seeking to limit social bot activity and bots that are constantly upgraded by their authors to possess new techniques for evading detection. Despite years of research, bots are as active and prevalent today as they have ever been.

The review of social bot literature reveals that crowd reactions (i.e., replies) to social bots have typically gone unused. This research explores whether crowd reactions are useful as a form of crowd-sourced data labels for augmenting more traditional approaches for social bot detection. However, not all crowd reactions are of equal credibility. In light of this, how can crowd reactions be weighed for the social bot detection task? Does speech act theory provide a viable perspective for how to capture the intention of individuals reacting to social bot-generated content?

A secondary focus of this research is to measure the real-world impact that social bots have on users. How prevalent are social bots and how do real users interact with them? By scrutinizing discussion threads between humans and bots, these questions can be computationally explored using data sourced from real-world interactions. These questions have generally been left unexplored in previous work. By understanding the realized impact bots have on real users and discussion, the social bot problem can be quantified. As a result, informed managerial decisions can be made by platform owners, for example, investment in antibot technology and other content management policies.

3.1. Research Design

A research framework (Figure 1) was developed to guide the development of a social bot detection system. First, social media data are collected using publicly available means. Next, interactions between humans and social bots are extracted, primarily including reply-to threads. Next, methods borrowed from computational linguistics and machine learning are used to extract signals from human and social bot messages that may be useful for the bot detection task. Speech act theory as described in Searle (1969) is the speech act framework used in this research given its recent use in other computational and IS research (Abbasi et al. 2018). Several features are then generated from each account's messages, including topic, sentiment, speech act, semantic features, and temporal features. These features are generated independently and later combined for predicting bot activity rather than having one model jointly construct these features simultaneously. Last, experimentation and evaluation of the bot detection system is performed.

4.1. Data Collection

4.1.1. Social Media Platform Description. Data for this study were collected from Reddit.com, one of the world's most popular social media websites. Overall, the Reddit context provides an interesting opportunity to study and learn from multiple interactions between humans and social bots. Data sourced from Reddit also represent human-bot social interactions as they occur naturally in the real world rather than controlled settings. Millions of new messages are posted by users daily on the Reddit platform. Reddit also enjoys an international community that can participate within various "subreddits" (i.e., topic-specific discussion boards) across the platform. Subreddits can be focused on any topics related to news, sports, politics, science, and much more. Any one user can also create their own subreddit and maintain full moderator rights over it.

Despite there being multiple subreddits for many topics and purposes, the mechanisms for users to converse and interact with one another remains consistent across the platform (see Online Appendix A1 for a sample conversation involving a bot). There is a quantifiable impact that social bots possess over online discourse that can be measured by identifying bot-generated messages scrutinizing how much interaction they garner among real users. Although many social bots are benign in nature, there are numerous instances of malicious social bots attempting to influence discourse. Given its size and popularity, Reddit has become a frequent target for cyber-adversaries wishing to manipulate and exploit unsuspecting users. There is significant social bot activity, and many users unknowingly consume and interact with bot-generated content. Many subreddit moderators, who are typically volunteers, struggle to combat the propagation of bot content. For instance, Figure 2 contains an example of a Reddit post where a moderator of a moderately sized subreddit cautions fellow users to remain vigilant against increased bot activity. Despite their efforts and having already banned some known bot accounts, social bots continue to attack the subreddit.

Human moderators are capable of detecting social bot activity as exemplified in Figure 2, but manual moderation efforts cannot be scaled to handle the volume of increased bot activity or to handle bot activity over sustained lengths of time (Kiene et al. 2019). As a result, Reddit users have developed a community-maintained list of bot accounts that subreddit moderators can proactively monitor. Data were obtained from the

Figure 1. (Color online) Research Framework

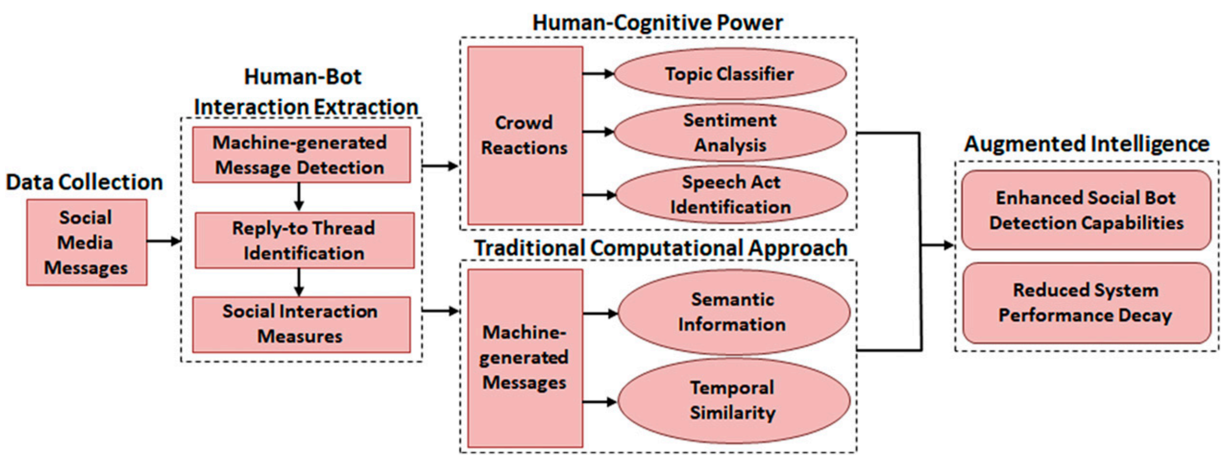
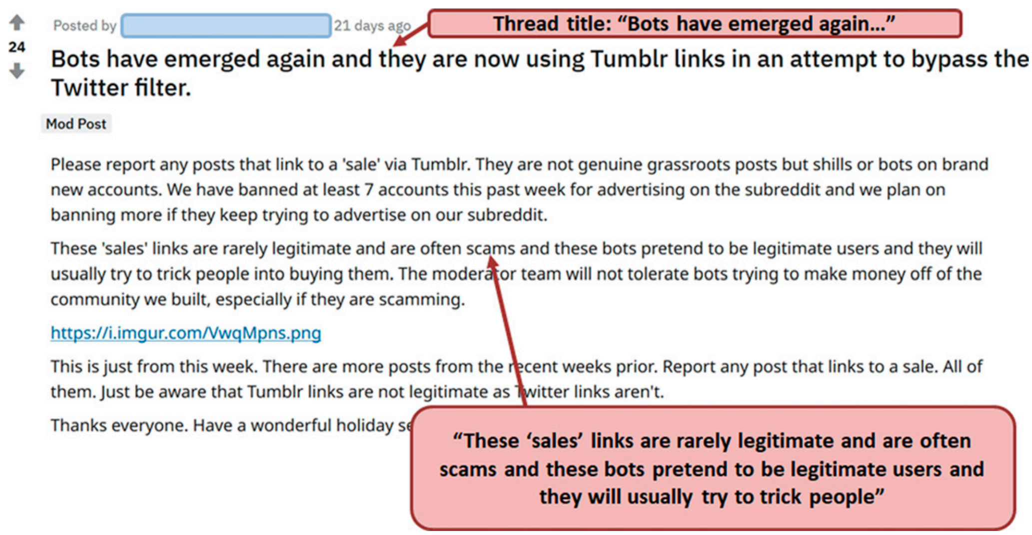


Figure 2. (Color online) Subreddit Moderator Reaction to Increasing Social Bot Scam Activity



Reddit community called *r/BotWatchman*. An alternative subreddit that appears to serve the same purpose is *r/BotWatch*. Updated bot lists seem to be periodically shared by participants. However, precaution is necessary as the open-source nature of such bot lists implies that there is nothing preventing users from submitting false reports.

4.1.2. Collection Process. Reddit.com provides an official, publicly accessible Application Programming Interface (API) that can be used to retrieve and submit user account and conversational data. In this study, the Reddit public API is used to collect conversational data relevant to social bot accounts featured in the crowdsourced data set.[1] There are just over 4,000 accounts featured in the data set; however, to combat against potential false reports, only listed accounts with 10 or more reports are included in this study. This reduces the number of bot accounts from more than 4,000 to 777. The filtered data set results in a higher quality ground truth representation testbed to experiment with. These 777 accounts are what we consider our ground truth data set of social bot accounts. After identifying this set of 777 social bots, we collected the entirety of every conversational thread that featured their participation. This includes analysis of human reaction to bot-generated content and the extent of visibility of bot content among human-generated messages.

4.1.3. Human–Bot Interaction Extraction. To effectively combine human and machine cognitive power, it is necessary to distill the conversational threads down to the specific interactions between bots and humans. These interactions are naturally captured in reply-to relationships between a social bot's message and the human replies it generates. Table 3 presents a summary of the data used for this research. Social media data collected for this work span the years 2016–2018. First, all messages pertaining to the 777 identified social bot accounts are extracted, totaling 146,639 bot-generated messages. Reddit threads contain two types of messages: (1) root-level comments that typically serve as subtopics or conversation starters and (2) replies to these root-level comments. The extracted 146,639 messages are of this nature rather than replies to other humans. Next, all replies written in direct response to the 146,639 social bot messages are collected. This totals 58,070 replies. Among these reply messages, there were no instances of one of the 777 known social bot accounts replying to another bot-generated message. Thus, we assume all reactions to bot-generated messages are indeed written by humans and not by other social bots. Furthermore, we do not collect all account data for those accounts that generate a reply to a social bot message. Because the focus of interest here is to observe human reactions to bot-generated content, only instances of human–bot interaction are collected; collection of all account activity by those accounts replying to bot-generated content would result in capturing many human–human interactions, which is not a focus here.

However, this work will also include experiments that will attempt to scrutinize bot accounts from genuine, human-operated accounts. Thus, it is necessary to collect all activity related to a sample of human accounts. Literature estimates that bot activity accounts for about 5%–8% of all social media activity (de Lima Salge and Berente 2017). We assume the 146,639 collected bot messages are representative of the 6.5% (the median of 5%–8%) total of social bot activity and that

Table 3. Data Summary

Data category	Number of accounts	Number of messages	Minimum word length	Maximum word length	Average word length
Bot accounts	777	146,639	1	1,643	51
Replies to bot accounts	—	58,070	1	893	9
Nonbot accounts	195,291	2,256,018	1	5,691	44
Replies to nonbot accounts	—	793,398	1	1,611	23
Human messages with bot replies	—	82,266	1	5,261	19

human messages account for the other 93.5% of data. To construct the testbed of human messages, we collect all account activity for nonbot accounts until the 93.5% threshold is met, resulting in 2,256,018 messages collected across 195,291 accounts. Similarly, all reply data to human messages are collected, so that there can be a comparison between replies to bot-generated and human-generated messages, totaling 793,398 messages. Finally, we also capture all human messages identified to have a reply from one of the known 777 bot accounts. Collecting such data allows for this study to explore human–bot interaction. Furthermore, one attribute of this data set is that there is a temporal order in which social media messages are posted by users. Model training and testing should take this into account to avoid feature leakage issues (Melis et al. 2019). For example, feature leakage would occur if a more recently generated datapoint is selected for model training and subsequently used to predict much older records. For this reason, we select an 80/20 split for model training and testing of our data set, leaving 622 accounts for model training and 155 for model testing. Additionally, if any comments made by the 622 training accounts receive message replies within the temporal scope in which the testing accounts are active, those replies are dropped from the data set to ensure no feature leakage occurs. Data used to operationalize features (e.g., speech acts) are sourced exclusively from training set accounts. Section 4.2.3 elaborates a bit further on this procedure.

4.2. Human-Cognitive Power

4.2.1. Vector Representation of Social Media Text Data. The primary innovation proposed in this study is to augment traditional approaches for social bot detection by incorporating human-cognitive power embedded within social media messages. The research design must account for how to process and extract meaningful signals from social media text data, particularly when evaluating the credibility of crowd-sourced labeling of potential social bot activity. Therefore, it is necessary to vectorize social media messages to make them suitable for large-scale analysis.

There is a very active stream of research within the NLP community to advance capabilities for learning vector representations of text at the word-level, referred to as word embeddings (Mikolov et al. 2013, Levy and Goldberg 2014, Reimers et al. 2019). Several subsequent studies extended and improved the performance of word embedding generation (Subramanian et al. 2019). Alternative techniques that use sentence-level embeddings rather than word level (Liu et al. 2015, Cer et al. 2018, Reimers et al. 2019) typically include transformers (e.g., Bidirectional Encoder Representations from Transformers (BERT)) and often outperform word-level embeddings.

4.2.2. Transformers. Transformers (i.e., BERT, Generative Pre-trained Transformer (GPT)-3) represent the latest advancements in NLP methodology and are thus used in the curation of the described system (Reimers et al. 2019, Song et al. 2019). Although a thorough review of transformers is outside the scope of this article, it is important to note that one of their perceived benefits is the capacity to accept a theoretical infinitely long input sequence of text, process it, and produce a relevant output sequence. This capability can be contrasted against RNNs (e.g., Long Short-term Memory (LSTM)), which are also popular for sequence mining applications, but suffer from a vanishing gradient problem as the input sequence becomes lengthier (Sundermeyer et al. 2012, Wang 2017). That is, the information from the beginning of an input sequence diminishes as the model iterates further along through the sequence, thus resulting in overall performance loss. In the context of social media, it is not unexpected to encounter lengthy user-generated text, which may render RNNs to be less effectual than transformer methods; this has been demonstrated as such in various benchmark experiments (Sun et al. 2019). Thus, this research elects to use transformers to develop the described system.

To use transformers, this study makes use of the Hugging Face Transformers library, which has emerged as one of the most popular implementations of BERT (Wolf et al. 2020). The transformers library provides a BERT model with 109,483,778 parameters that was pretrained over hundreds of millions of tokens. This model is adopted by this study. Furthermore, the model is coupled with a dropout layer to prevent overfitting and a final dense layer for classification tasks.

4.2.3. Topic Classification. To implement the human cognitive component of the research framework, it is necessary to develop the capability to determine whether human replies to social bots are indeed identifying bot activity. This can be operationalized through machine learning classification of reply messages where each message will be evaluated on whether it is related to discussion of social bots, resulting in binary classification between relevant and nonrelevant posts. To develop a training data set, two student annotators generated a labeled set of 2,000 reply messages, with 1,000 messages assigned to each class to maintain balanced class size during model learning. Messages were sampled at random from an identified subset of 622 training accounts that include instances where bot-like activity was indeed identified in other users' replies. Not all bot messages yielded replies where bot activity was identified; many users replied to the bot accounts with genuine responses, assuming a real conversation.

The model is trained with balanced classes but is ultimately used to predict labels of real-world social media data where class balance cannot be ensured. An alternate model configuration is to train the model on imbalanced class sizes that more closely represent real-world conditions. Numerous works have investigated the performance differences between the two approaches, with evidence pointing toward learning from imbalanced data typically results in deteriorated performance (Han et al. 2005, He and Garcia 2009). Such findings have given rise to studies focused on how to best learn from imbalanced data, often through random oversampling of data points assigned to smaller classes (Fernández et al. 2018). The approach taken by this study aligns with literature.

A 10-fold evaluation was then performed on the labeled data set. The annotated data were then used to train the classification layer of the adopted BERT model. Other traditional classification models were also selected for benchmarking and evaluation purposes, including support vector machines (SVM), a traditional RNN, LSTM, and bidirectional LSTM. Results can be viewed in Table 4. It should be noted that because class sizes are balanced for model training, the macro- and microaverages are the same in this context as no single class should be weighted more heavily. The BERT model performs best overall for all three classification tasks. This outcome is likely because of its ability to use long input sequences with no information lost to vanishing gradients that can impact other methods, such as RNN-based techniques.

4.2.4. Sentiment Analysis. Sentiment is another factor to consider when assessing human replies to potential bot-generated messages. Human replies to perceived bot-like activity may be polarized in sentiment, and thus sentiment may be a valuable feature that helps reliably unveil social bots. For example, some social bots may elicit negative sentiment if they are discovered as attempting to subvert and misinform users. The two annotators generated a labeled set of 3,000 messages, with 1,000 messages being labeled as positive, neutral, or negative sentiment. In this analysis, messages were also sampled from only replies to known bots. However, we opt to enforce this constraint to maintain consistency with our sampling strategy for the topic analysis than for any other concerns. We reason that semantic information pertaining to sentiment is generalizable, and thus constraining the selection of training data is not a concern here. The annotators reached 97% reliability. Once again, the classifier is trained on balanced data but tested on imbalanced, real-world data. Furthermore, just as with topic classification, several models were evaluated for predicting sentiment. Results are displayed in Table 4.

4.2.5. Speech Act Identification. Speech acts can reveal information regarding a human's credibility in their judgement of whether they have identified a bot. For example, a message in which a user declares that they have found a bot contains a higher level of certainty than a message in which one user questions others whether they have identified bot-generated content. Not all human-curated replies will be useful for augmenting the bot-detection task, nor should they be weighed equally. Speech acts may help clarify which human replies should bias model learning the most by the selected classification methods.

Two annotators labeled social media messages until they were able to identify a total of 500 messages for each of the five speech acts described in speech act theory (Searle 1969). Much like the topic and sentiment analyses, only human replies to known bots were sampled for the training set to maintain consistency across the three analyses. The resulting training set consisted of 2,500 messages across all speech acts. The labeling task was to assign the most appropriate speech act label to a given message, and thus each message was given only one speech act label. The annotators achieved 90% interrater reliability on this task. Examples of labeled sentences from our data set can be found in Online Appendix A2.

Evaluation was performed using multiple models with 10-fold validation similar to the topic and sentiment classification tasks (Table 4). BERT was once again the top performer. The classifier was then used to label the remaining reply messages in the data set. Among replies to bots, the expressive category was the most prevalent account for more than half of speech acts. Assertive speech acts were the next

Table 4. Results of Topic Classification, Sentiment Analysis, and Speech Act Classification

	Precision (average)	Recall (average)	F_1	AUC	Standard error
Topic classification					
BERT	*0.967*	*0.951*	*0.959*	*0.957*	*0.001*
BiLSTM	0.964	0.943	0.953	0.953	0.001
LSTM	0.938	0.926	0.932	0.941	0.001
RNN	0.881	0.886	0.883	0.928	0.001
SVM	0.788	0.813	0.800	0.879	0.003
Sentiment classification					
BERT	*0.928*	*0.900*	*0.914*	*0.902*	*0.002*
BiLSTM	0.894	0.873	0.883	0.881	0.002
LSTM	0.868	0.856	0.862	0.878	0.003
RNN	0.812	0.816	0.814	0.852	0.003
SVM	0.712	0.743	0.727	0.744	0.005
Speech act classification					
BERT	*0.862*	*0.869*	*0.865*	*0.8495*	*0.004*
BiLSTM	0.839	0.854	0.846	0.842	0.004
LSTM	0.776	0.767	0.771	0.835	0.004
RNN	0.726	0.735	0.730	0.781	0.005
SVM	0.561	0.588	0.574	0.657	0.007

Note. The italicized entries are the best performing models for each experiment that was run.

common (~18% of data set). Declarative speech acts and directive speech acts appeared less frequently (each ~10%). Commissive speech acts toward bots appeared the least frequently. Overall, distinguishable speech act patterns can be observed when examining replies that successfully call out bot activity. However, among replies to social bots that fail to identify algorithmically generated content, speech act use was not statistically different than replies to nonbots. In fact, replies to bots that did not make notice of bot activity and replies to nonbots shared more semantic similarities and are overall more linguistically diverse than replies to bots that successfully identified social bot activity.

4.3. Traditional Computational Approach

Traditional machine learning approaches for text-based social bot detection primarily consist of semantic analysis of bot messages and temporal analysis of message similarity (Hegelich and Janetzko 2016, Chavoshi et al. 2017). This study uses both strategies as they have demonstrated to yield high performance in recent publications within the bot detection research stream. Messages used for this analysis can be sampled from the set of known social bots identified by the crowd. The semantic information these accounts post can be used to further assess potential bot activity.

4.3.1. Semantic Information. Semantic information may be important for unveiling attack strategies or linguistic trends in messages posted by social bots. For example, recall that Figure 2 showcased how a subreddit moderator frequently encounters and has trouble controlling bots that use a strategy of trying to scam users through advertisement of false "sales." Semantic information is operationalized at the account level for the bot and nonbot accounts. Account-level embeddings are generated by concatenating all messages an account posts into a single input that is fed into a BERT model used to generate an embedding. Because transformers can theoretically accept infinitely long inputs, this technique seemed to be the most efficient way to use BERT for representing all semantic information posted by a singular account.

4.3.2. Temporal Similarity. Temporal similarity will help identify whether an account continuously posts the same or similar messages over time, indicating bot-like activity. Recall that literature indicates that social bots can maintain highly synchronous activity over time (Chavoshi et al. 2017). Cosine similarity can be computed between pairs of messages posted by a single account to understand overlapping semantics across messages. The cosine similarity scores can then be averaged and used to represent the average similarity of an account's posted messages; exhibiting high similarity between messages may be indicative of bot activity. In this research, we take the average similarity across all messages posted by an account.

4.4. Human-Machine Symbiosis

4.4.1. Social Bot Detection Classifier. After extracting the human-cognitive power embedded within reply messages and implementing traditional social bot detection methods built on semantic information, it is possible to evaluate the performance of intelligence augmentation for the bot-detection task. A feature matrix is constructed (Table 5) and accounts for (1) the proportion of replies identifying bot activity (i.e., Bot Topic), (2) the proportion of replies classified as positive, negative, and neutral, (3) the five speech acts

Table 5. Classifier Input Matrix Example

	Bot topic	Positive	Neutral	Negative	SA1	SA2	SA3	SA4	SA5	Temporal similarity	V_s
X_1	84%	17%	32%	51%	20%	10%	40%	15%	15%	78%	...
X_2	0%	40%	50%	10%	12%	4%	6%	18%	60%	12%	…
…	…	…	…	…	…	…	…	…	…	…	…
X_n	64%	22%	65%	13%	4%	25%	16%	35%	20%	52%	…

used in this study (i.e., SA1, SA2, SA3, SA4, and SA5), (4) a temporal similarity measure, and (5) semantic information of each account encoded as BERT embeddings V_s. All text-based features used in this final classification were generated using BERT as previously described in Table 4 because it performed best among the models evaluated.

Several classification models were selected for trial in this study, including (1) random forest, (2) decision tree, (3) logistic regression, (4) SVM, and (5) naïve Bayes. The experiment is run in two configurations. For the baseline model, social bot detection is attempted across all bot accounts using only the traditional computational approaches, including semantic information and temporal similarity (Ferrara 2018). The second configuration of the experiment combines the traditional features with human cognitive features. In each experiment configuration, accounts are being classified based on whether they are a bot or not.

Overall, the augmented model provided considerable lift performance across all classification methods (Table 6). In this experiment, random forest yielded the best results and helped demonstrate that incorporating human-cognitive power appears to indeed boost the model's ability to correctly distinguish between social bots and genuine users. Furthermore, the human cognitive features boost the model's ability to identify a greater number of the social bot accounts in the data, thus reducing false negatives.

Results in Table 6 demonstrate that the augmented model yields better performance. However, within the augmented model, it is unclear if performance gains come solely from using reply messages or whether the speech acts contribute toward improving performance. Thus, the developed model was benchmarked using restricted feature sets. The analysis with reduced feature sets shows that the inclusion of speech acts provides a modest performance improvement (Table 7). However, when considering operationalizing on social media platforms where millions of messages are posted daily, even a slight bump in performance could impact thousands of bot detection cases that would have previously gone undetected. Online Appendix A3 contains a deeper look at how each speech act performs individually.

4.4.2. Time-to-Detection. To further demonstrate that the proposed framework is pragmatic and can be deployed in live environments, a real-world detection scenario is simulated. In the prior experiment, the data set was split into a training and testing set based on the temporal occurrence of bot accounts; older accounts were used for model training, whereas more recent accounts were withheld for testing to avoid feature leakage. This experiment schema can also be exploited to operationalize a simulation of how real-world detection could occur. Bot accounts in the testing set are evaluated at three intervals, once when the account has posted 10 messages, once at 25 messages, and again at 50 messages. These intervals were chosen to assess how rapidly the framework can respond to new social bots. The percent of known bots detected at each message-interval is featured in Table 8. Both the traditional and augmented models are evaluated. Many social bots can

Table 6. Social Bot Detection Classification Results

	Precision (bot class)	Recall (bot class)	Macro-F_1	Micro-F_1	AUC	Standard error
Traditional approach						
Random forest	*0.478*	*0.354*	*0.407*	*0.401*	*0.418*	*0.008*
Decision tree	0.467	0.303	0.368	0.361	0.373	0.008
Logistic regression	0.459	0.266	0.337	0.335	0.354	0.009
SVM	0.355	0.253	0.295	0.268	0.317	0.010
Naïve Bayes	0.347	0.245	0.287	0.270	0.303	0.011
Augmented model						
Random forest	*0.843*	*0.770*	*0.805*	0.781	*0.817*	*0.004*
Decision tree	0.821	0.722	0.768	0.742	0.790	0.004
Logistic regression	0.796	0.706	0.748	0.736	0.765	0.005
SVM	0.705	0.666	0.685	0.666	0.703	0.005
Naïve Bayes	0.679	0.638	0.658	0.646	0.664	0.006

Note. The italicized entries are the best performing models for each experiment that was run.

Table 7. Ablation Analysis for Feature Set Comparison

	Precision (bot class)	Recall (bot class)	Macro-F_1	Micro-F_1	AUC	Standard error
Model excluding topic						
Random forest	*0.763*	*0.720*	*0.745*	*0.742*	*0.758*	*0.007*
Decision tree	0.735	0.712	0.715	0.695	0.722	0.008
Logistic regression	0.729	0.682	0.706	0.679	0.714	0.008
Naïve Bayes	0.722	0.644	0.681	0.671	0.692	0.011
SVM	0.706	0.607	0.653	0.648	0.670	0.014
Model excluding sentiment						
Random forest	*0.854*	*0.764*	*0.806*	*0.785*	*0.819*	*0.004*
Decision tree	0.820	0.720	0.767	0.745	0.782	0.004
Logistic regression	0.796	0.696	0.742	0.721	0.749	0.005
Naïve Bayes	0.742	0.671	0.705	0.692	0.720	0.005
SVM	0.739	0.661	0.698	0.685	0.700	0.006
Model excluding speech acts						
Random forest	*0.829*	*0.726*	*0.774*	*0.754*	*0.791*	*0.007*
Decision tree	0.797	0.712	0.752	0.748	0.765	0.007
Logistic regression	0.757	0.701	0.728	0.712	0.740	0.008
Naïve Bayes	0.734	0.657	0.710	0.695	*0.718*	0.009
SVM	0.711	0.520	0.696	0.672	0.705	0.009

Note. The italicized entries are the best performing models for each experiment that was run.

be detected with just a small sample of their conversational activity, and over time, the likelihood of the crowd identifying bot-like activity will increase. When considering the scale of a platform like Reddit, a few percentage points of improvement can result in thousands of additional bots being identified.

Additionally, classification performance is evaluated as a function of how many crowd labels have been generated against suspected bot accounts. Classifier efficacy is evaluated when a single crowd label is generated and when 5 and 10 labels are generated by users (Table 9). The performance of both the traditional model and augmented models are compared. Classification tests are triggered for both models simultaneously on the occurrence of crowd-generated labels; however, the traditional model does not actually incorporate the crowd labels as an input feature for classification. Results demonstrate that the augmented model is greatly assisted by its ability to incorporate crowd labels into its classification decisions.

4.4.3. System Performance Against Advancing Bots.

There are two primary reasons bot detection performance increases as more content is posted by suspected accounts. First, reliance of traditional bot detection features, such as message similarity, can become indicative of bot activity as more evidence is presented. Many bots will likely reveal themselves in time by repeating messages or focusing on singular topics for extended lengths of time not characteristic of human activity. Second, as an account's post volume increases over time, the likelihood that it has received replies from others also increases. These reply messages are used in the framework presented by this research.

Despite the demonstrated performance, a remaining concern is that social bots will become more advanced over time and develop capabilities to evade existing detection methods. To further explore the practicality of the proposed system in real-world contexts, the proposed bot detection classifier is tested against a set of more recently identified social bots. Because the social bots are more recently active, it is possible they possess more advanced capabilities and ability to evade detection.

Recall that the data set described in Table 3 and used for the main experiment in this paper spanned from 2016 to 2018. An updated list of crowd reports on potential bot activity from 2019 was identified. The same data processing procedures used in the main experiment were also applied here. Primarily, the testbed was restricted to only include data pertaining to social bots with a minimum of at least 10 crowd reports, indicating a higher level of credibility for establishing ground truth. Furthermore, some social bots self-identified themselves as bots in this data set, either in their username or posted messages. These accounts were removed through a mixture of keyword searches and manual vetting. Finally, to establish

Table 8. Results of Example Simulation

	Ten messages	Twenty-five messages	Fifty messages
Traditional model	52%	58%	64%
Augmented model	53%	71%	81%

Table 9. Results of Example Simulation

	One crowd label	Five crowd labels	Ten crowd labels
Traditional model	48%	53%	59%
Augmented model	53%	68%	78%

a meaningful time gap between the main data set used in this research and the data used for this secondary experiment, only social bot accounts created in the second half of 2019 are used. The final data set consists of 260 social bots that were created and active on Reddit from June 2019 to March 2020. During this time span, bots posted 23,574 messages, which subsequently resulted in 5,592 replies. A collection of 362,677 messages posted across Reddit were also randomly collected form this same time window to serve as a supposed nonbot class.

The goal of this secondary experiment is to test the ability for the system described by this research to successfully detect social bots that may be continuously advancing in the wild (Table 10). Thus, models trained previously in this research are applied toward detection of this newly identified subset of social bots. The methodological procedures described previously are applied to this new subset of data, resulting in a feature matrix resembling Table 5. With the new data, the augmented model does not experience a significant performance loss. The lack of consideration toward users replies in the traditional approach seems to contribute to increased performance loss compared with the augmented model. Overall, this experiment demonstrates that incorporating user replies in conjunction with speech acts may help prevent some performance degradation over time. Performance could be further improved by retraining the detection system with newer data.

4.4.4. Considering Other Benchmarks. We attempted some direct benchmarking against methods in literature. The most direct study to benchmark against is Garcia-Silva et al. (2019), as this study uses various advanced models, including BERT, to develop Twitter bot detection capabilities. In their study, they use BERT to achieve an F_1 score of 0.8388 on the Twitter data set of Cresci et al. (2017). Garcia-Silva et al. use BERT base released for use with TensorFlow, but the same model is made available in the HuggingFace Transformers library used in our study; performance appears to be largely comparable. Garcia-Silva et al. thoroughly benchmark the performance of BERT against RNNs (BiLSTM) with embeddings constructed using a variety of state-of-the-art methods, such as FastText and Embeddings from Language Models (ELMo). BERT is shown to be the leading model. We attempted to apply our own model against the same data set of Cresci et al. (2017) to produce benchmark results. However, because the data set of Cresci et al. (2017) does not contain any message reply data, it is not possible to benchmark any of the crowd reaction features we use in our study. However, what is possible is to validate our BERT model and some of the other more traditional features used in this study by comparing our results with those in Garcia-Silva et al. (2019). Overall, we find comparable results for BERT base, whereas our study slightly outperforms Garcia-Silva et al. when considering two additional features that are possible to implement (i.e., temporal similarity and topic detection). See Online Appendix A6. These results are similar to what we observe on the Reddit data set; adding in the two features can enhance performance, and then incorporating the crowd reaction features extends performance even more. The benchmark comparisons seem to imply that the additional features do indeed enhance performance over baseline models, and if Cresci et al. (2017) possessed reply data, we expect our feature set to perform even better.

Similarly, we explored the possibility of using existing social bot detection methods from literature and applying them to our own data set. In this case, we did not necessarily expect to find other methods using replies as a feature (we had found no evidence such papers exist), but rather, we searched for competing models that do not use replies yet still yield good performance. Comparing against such methods would help better substantiate the role played by speech acts and message replies. For example, we look toward Varol et al. (2017) because it provides a comprehensive list of features useful for social bot detection. However, we found that many features used simply do not crossover to Reddit, and thus we could only partially implement the model from Varol et al. Specifically, of the 63 feature categories originally used in the study (table 1 in Varol et al. 2017), we could only reasonably implement about 33 features using data that Reddit makes publicly available. Many of the features could not be generalized away from the Twitter platform (e.g., number of retweets per hour), or they may rely on Twitter's networking features (e.g., number of followers) that could translate effectively to some number of social platforms but unfortunately not Reddit because of its lack of public "friend" or "follower" features. Many other bot detection papers appear to present similar challenges for application to our data set. We decided to forego such analyses as

Table 10. Results of Example Simulation

	Precision (bot class)	Recall (bot class)	Percentage change in precision compared with original experiment	Percentage change in recall compared with original experiment
Traditional model				
Random forest	0.436	0.314	−8.78%	−11.299%
Decision tree	0.419	0.271	−10.278%	−10.561%
Logistic regression	0.422	0.243	−8.061%	−8.647%
Naïve Bayes	0.308	0.225	−13.239%	−11.067%
SVM	0.282	0.206	−18.732%	−15.918%
Augmented model				
Random forest	0.814	0.742	−3.440%	−3.636%
Decision tree	0.782	0.683	−4.750%	−5.402%
Logistic regression	0.759	0.671	−4.648%	−4.958%
Naïve Bayes	0.656	0.633	−6.950%	−4.955%
SVM	0.650	0.592	−4.271%	−7.524%

they would not be representative of the original models and would not give us meaningful conclusions.

5. Discussion and Conclusion

5.1. Overview

Social bots have the power to frame online conversation and harm the health of online discourse. To address this high-impact issue, this research develops a framework for social bot detection that uses crowd labeling and crowd reactions toward content curated by social bots. The developed framework accomplishes this capacity through a combination of computational methods and linguistic theory. Our work makes several contributions to social bot detection and crowd-sourced labeling literature. Specifically, identifying user replies to social bot content and subsequently categorizing them by their speech act to measure their credibility for social bot detection is a novel approach not previously taken in literature. Although social bots successfully deceive many social media users, other users have become more aware of bot activity and may identify it within social media comments. This is a form of natural crowd-sourced labeling; however, not all labels are equally credible. By using speech act theory to characterize the different intentions users have when replying to social bot content, this study demonstrates improvements in social bot detection systems performance.

One of the core strengths of the proposed framework is its capability to be deployed to the real world. Many social media platforms already make use of bot detection systems that incorporate the traditional approach described in this manuscript. Generating the features described in this research from message replies is a natural extension to these already existing systems. Furthermore, in contexts where a bot detection system must be built from scratch, the described framework makes use of conventional techniques and demonstrates improved performance gains by also implementing the augmented model. It is quite feasible for the research framework to be deployed in real-world contexts and successfully reduce bot-generated content.

This work and other social bot research is becoming increasingly important to society as problems like "fake news" and manipulation of social media are among the most challenging issues that threaten social and economic progress. Since the 2016 U.S. elections, the conversational health of online social media has come into question, with social media platforms and other online communities facing increasing scrutiny to dampen the negative impacts and widespread reach that cyber-adversaries within online discourse (Broniatowski et al. 2018). For example, in the United States, there is increasing political debate regarding the role and responsibilities that social media platforms should assume in vetting content posted onto their platform. Recent focus in the United States has been placed on Section 230 of the 1996 Communications Decency Act, which affords social media platforms and other Internet-based businesses a legal shield against illicit content that users may independently upload onto such platforms. Revoking Section 230 would make companies liable to monitor and remove user submissions deemed inappropriate, which is an issue of focus in a 2020 Executive Order issued by the U.S. president.

Given that much social media manipulation is automated using social bots, investing effort into advancing social bot techniques is pragmatic and possesses real-world value. The feature sets and methodological approaches considered in this study contribute to the growing body of literature that aims to solve this high-impact problem of great importance to society. Additionally, even with the contributions of this paper, the social bot problem is not "solved"; as Yang et al. (2019) stress, there is a need for continuous development of new feature sets and methodology to combat against continuously advancing social bots.

5.2. Limitations

The research design described in this study is not without its limitations. The reliance of crowd labeling for social bot detection is, in some measure, a strategy that permits bots to act and persist for some length of time. This means the bot could theoretically cause some harm to the platform or users before it is discovered. Thus, the best this method could achieve is a delayed response to social bot detection. However, no existing method can detect 100% of social bots before they begin interacting with target platforms, and therefore, there is still value to the approach described in this research.

A second limitation is that this approach relies on social bots that users could identify but does not address social bots that can successfully operate without human detection. Because the described approach relies on crowd labeling, its performance is somewhat tethered to the general user's capability for identifying social bot activity. Although this approach may not detect the most advanced social bots at a given time, the reliance on crowd labeling means that the performance of the system will naturally improve over time as users also improve their capabilities to identify bots. Similarly, the reliance of crowd labeling may bias the data set to some forms of social bots instead of others.

One consequence of reliance on speech act categorization is that it may introduce bias depending on personality of the individuals included within the data set. For example, a person might be stoic by nature and exhibit less expressive speech acts. In these cases, speech act use may become a function of personality rather than confidence toward the bot detection task. This paper does not explicitly address this issue. Using many crowd-sourced labels may help reduce potential bias from any single individual.

Previous research has demonstrated that some social bots can interact with one another to make each other appear more legitimate (Abu-El-Rub and Mueen 2019). Such activity generally requires the bot owner(s) to maintain a network of more advanced social bots that can act in a coordinated fashion; our framework can still help detect such social bots if human users detect suspicious activity.

Another limitation is that bots may eventually practice a strategy of posting false claims of suspected bot activity. This could lead to false positives and disrupt the efficacy of the proposed bot detection system. Because we lack ground truth regarding what bots may be practicing this strategy, we are not able to investigate the issue in this study. The data we use in this study include suspected bot accounts reported by multiple users on the Reddit platform. We assume accounts with a high number of reports to be ground truth, and among these accounts, we do not observe any of them implementing this strategy. However, further exploration is necessary to investigate the efficacy of this strategy for detection avoidance.

5.3. Practical Insights

Along with a social bot detection system, this research offers other knowledge to security practitioners and platforms combatting social bots. The data set offers some insights into the actual impact that social bots have within online discourse. For example, the number of interactions between social bots and humans can be quantified and used to understand the reach social bots have on real users. Online Appendix A4 summarizes descriptive statistics that help further illuminate social interactions between bots and humans as observed within the crowd-sourced data set. In general, social bots create messages that are less harmonious in terms of topic and sentiment compared with overall conversation. Furthermore, Online Appendix A5 provides some metadata on users that replied to bot posts and contrasts differences between those who identified bot activity and those who did not. Interestingly, most bots were detected by distinct users rather than a subset of the same users being responsible for most reports. This lends credibility to the potential contributions of crowd-sourced approaches.

The proposed system framework is intended to be feasibly deployed by any social platform seeking to reduce bot-generated content. First, a social platform must train an initial model for bot detection. The data used in this study can help platforms train some baseline model for use, or alternatively, social platforms can label their own data sets with more contextually relevant data. Social platforms would benefit from labeling their own data and crowd reactions to suspected bot messages. For example, a bot detection model trained on Twitter data would likely not perform well against Reddit data, given that the length of Twitter and Reddit posts tend to greatly differ, impacting the semantic information users share on these platforms. In either case, platforms must deploy a pretrained model to avoid a cold start.

After an initial model is trained, it can be deployed for use. The model could be used to detect potential bot accounts that begin operating during any time after the activity captured in the training data. Execution of the model to scrutinize potential bot accounts could happen at variable intervals depending on context and infrastructure capabilities. For example, many modern web platforms today make use of scalable, on-demand infrastructure that could allow for content analyses to be executed at the time of user submission. Conversely, under computational resource constraints, content analyses can be performed in scheduled intervals with appropriate resource allocation. This could occur in intervals that account for several hours of content at a time, or even analysis at a daily level. Delaying analysis further than at a daily level could risk allowing bot-generated content to proliferate and harm user experiences. Moderation decisions could be made with assistance of the proposed system.

Any real-world bot detection system would also need regular updating, as bot detection is a moving target, and the semantic information bots propagate will shift over time. It is important to update and retrain the model at regular intervals. Training samples can come from two inputs. First, bots that are discovered by the system can be recycled as inputs used to further train the system. This approach would be particularly helpful to detect bots focused on a general conversational topic but gradually shift their messaging regarding the topic; for example, social bots spreading political propaganda may focus on few topics and/or perspectives but evolve their messaging over time to align with contemporary real-world events. A second stream of training examples should ideally come from external sources not identified by the system itself. The system will have a bias of what it can and cannot identify, and thus external data sources are necessary to escape such bias. Platforms can use internal teams to manually scrutinize and label potential bot content or even use future crowd-sourced lists to further train their detection systems.

5.4. Contributions and Future Research

There are many contributions of this work to the bot detection research stream. As demonstrated in the literature review, the vast majority of extant bot detection literature is focused on Twitter or other microblogs focused on short-form social media and do not consider crowd reactions (likely because of reliance of the Cresci et al. (2017) benchmark data set that lacks reply messages). The application of existing models and techniques on longer-form social media, such as Reddit, is unknown. Furthermore, the utilization of crowd reactions has been completely absent from literature up until this point.

It is popular in the bot detection community to use the Twitter data set of Cresci et al. (2017) for benchmarking. Given this, our data set from the Reddit platform represents a novel context that advances bot detection research in a new direction. Future studies in this research stream should seek to use crowd reactions wherever possible, as their efficacy for bot detection is well demonstrated in this study. Furthermore, we demonstrate how speech acts may be a vehicle for assessing the certainty of different crowd reactions. To the best of the authors' knowledge, this research represents the first instance in which speech act theory was used for augmenting bot detection systems with crowd-generated labels.

This study lies at the intersection of combining human and machine cognitive power. The system instantiated in this work uses state-of-the-art machine learning techniques and novel perspectives in the bot detection literature stream to solve a problem of growing interest in both academia and industry. The need for more computational work of this nature is echoed in many recent calls for more design science work in IS journals (Agarwal and Dhar 2014, Goes 2014, Rai 2017). Additionally, this research lends itself to investigations of how real users identify social bots during their interactions on social media and how such interactions can be leveraged to more rapidly identify malicious bots.

Endnote

[1] All users who register on the Reddit platform agree to usage terms that states their data will be made available. To access the API, a free Reddit account must be registered. After registration, it is possible to generate an account-specific API key. This key can then be used to initiate a connection to the Reddit public API and its available services. This study abided by all Reddit data collection policies.

References

Abad ZSH, Butler GP, Thompson W, Lee J (2022) Crowdsourcing for machine learning in public health surveillance: Lessons learned from Amazon Mechanical Turk. *J. Medical Internet Res.* 24(1):e28749.

Abbasi A, Zhou Y, Deng S, Zhang P (2018) Text analytics to support sense-making in social media: A language-action perspective. *Management Inform. Systems Quart.* 42(2):427–464.

Abu-El-Rub N, Mueen A (2019) BotCamp: Bot-driven interactions in social campaigns. Liu L, White R, eds. *Proc. World Wide Web Conf.* (Association for Computing Machinery, New York), 2529–2535.

Agarwal R, Dhar V (2014) Big data, data science, and analytics: The opportunity and challenge for IS research. *Inform. Systems Res.* 25(3):443–448.

Alexandersson J, Buschbeck-Wolf B, Fujinami T, Maier E, Reithinger N, Schmitz B, Siegel M (1997) Dialogue acts in VerbMobil-2. Verbmobil report, German Research Center for Artificial Intelligence, Kaiserslautem, Germany.

Austin JL (1962) *How to Do Things with Words* (Oxford University Press, Oxford, UK).

Benjamin V, Valacich J, Chen H (2019) DICE-E: Darknet identification, collection, evaluation, with ethics. *Management Inform. Systems Quart.* 43(1):1–22.

Broniatowski DA, Jamison AM, Qi S, AlKulaib L, Chen T, Benton A, Quinn SC, et al. (2018) Weaponized health communication: Twitter bots and Russian trolls amplify the vaccine debate. *Amer. J. Public Health* 108(10):1378–1384.

Cai C, Li L, Zengi D (2017) Behavior enhanced deep bot detection in social media. Zheng X, Zhang H, Xing C, Wang A, Zhou L, Luo B, eds. *Proc. IEEE Internat. Conf. on Intelligence and Security Informatics* (IEEE, New York), 128–130.

Cer D, Yang Y, Kong SY, Hua N, Limtiaco N, John RS, Constant N, et al. (2018) Universal sentence encoder for English. *Proc. Conf. on Empirical Methods in Natural Language Processing: System Demonstrations*, 169–174.

Chang JC, Amershi S, Kamar E (2017) Revolt: Collaborative crowdsourcing for labeling machine learning datasets. Mark G, Fussell S, Lampe C, Schraefel MC, Hourcade JP, Appert C, Wigdor D, eds. *Proc. CHI Conf. on Human Factors in Comput. Systems* (Association for Computing Machinery, New York), 2334–2346.

Chavoshi N, Hosseinm H, Abdullah M (2017) DeBot: Twitter bot detection via warped correlation. Domeniconi C, ed. *Proc. IEEE Internat. Conf. on Data Mining* (IEEE, New York), 817–822.

Cresci S, Di Pietro R, Petrocchi M, Spognardi A, Tesconi M (2017) The paradigm-shift of social spambots: Evidence, theories, and tools for the arms race. Barrett R, Cummings R, Agichtein E, Gabrilovich E, eds. *Proc. 26th Internat. Conf. on World Wide Web Companion* (Association for Computing Machinery, New York), 963–972.

D'Andrade RG, Wish M (1985) Speech act theory in quantitative research on interpersonal behavior. *Discourse Processing* 8(2): 229–258.

Davis CA, Varol O, Ferrara E, Flammini A, Menczer F (2016) Botornot: A system to evaluate social bots. Bourdeau J, Hendler J, eds. *Proc. 25th Internat. Conf. Companion on World Wide Web* (Association for Computing Machinery, New York), 273–274.

de Lima Salge CA, Berente N (2017) Is that social bot behaving unethically? *Comm. ACM* 60(9):29–31.

de Moor A, Aakhus M (2006) Argumentation support: From technologies to tools. *Comm. ACM* 49(3):93–98.

Dong W, Liao S, Zhang Z (2018) Leveraging financial social media data for corporate fraud detection. *J. Management Inform. Systems* 35(2):461–487.

Dutta H, Kwon KH, Rao H (2018) A system for intergroup prejudice detection: The case of microblogging under terrorist attacks. *Decision Support Systems* 113:11–21.

Efthimion PG, Payne S, Proferes N (2018) Supervised machine learning bot detection techniques to identify social twitter bots. *SMU Data Sci. Rev.* 1(2):5.

Fernández A, Garcia S, Herrera F, Chawla NV (2018) SMOTE for learning from imbalanced data: Progress and challenges, marking the 15-year anniversary. *J. Artificial Intelligence Res.* 61:863–905.

Ferrara E (2018) Measuring social spam and the effect of bots on information diffusion in social media. Lehmann S, Ahn YY, eds. *Complex Spreading Phenomena in Social Systems* (Springer, Berlin), 229–255.

Ferrara E, Onur V, Davis C, Menczer F, Flammini A (2016) The rise of social bots. *Comm. ACM.* 59(7):96–104.

Ferraz Costa A, Yamaguchi Y, Juci Machado Traina A, Traina Jr C, Faloutsos C (2015) Rsc: Mining and modeling temporal activity in social media. Cao L, Zhang C, Joachims T, Webb G, Margineantu DD, Williams G, eds. *Proc. 21th ACM SIGKDD Internat. Conf. on Knowledge Discovery and Data Mining* (Association for Computing Machinery, New York), 269–278.

Garcia-Silva A, Berrio C, Gómez-Pérez JM (2019) An empirical study on pre-trained embeddings and language models for bot detection. Augenstein I, Gella S, Ruder S, Kann K, Can B, Welbl J, Conneau A, Ren X, Rei M, eds. *Proc. 4th Workshop on Representation Learn. for NLP* (Association for Computiational Linguistics, Stroudsburg, PA), 148–155.

Goes PB (2014) Design science research in top information systems journals. *MIS Quart.* 38(1):iii–viii.

Goodman ND, Frank MC (2016) Pragmatic language interpretation as probabilistic inference. *Trends Cognitive Sci.* 20(11):818–829.

Goodman ND, Stuhlmüller A (2013) Knowledge and implicature: Modeling language understanding as a social contagion. *Top. Cognitive Sci.* 5(1):173–184.

Han H, Wang W, Mao B (2005) Borderline-SMOTE: A new oversampling method in imbalanced data sets learning. Huang D-S, Zhang X-P, Huang G-B, eds. *Proc. Internat. Conf. on Intelligence Comput.* (Springer, New York), 878–887.

He H, Garcia A (2009) Learning from imbalanced data. *IEEE Trans. Knowledge Data Engrg.* 21(9):1263–1284.

He S, Rui H, Whinston AB (2017) Social media strategies in product-harm crises. *Inform. Systems Res.* 29(2):362–380.

Hegelich S, Janetzko D (2016) Are social bots on Twitter political actors? Empirical evidence from a Ukrainian social botnet. Gummadi KP, Strohmaier M, Gilbert E, Macy M, Wagner C, eds. *Proc. 10th Internat. AAAI Conf. on Web and Social Media* (AAAI, Palo Alto, CA), 579–582.

Heidari M, Jones JH (2020) Using bert to extract topic-independent sentiment features for social media bot detection. Vuong S, Chakrabarti S, Bradford P, Paul R, Rubenstein C, eds. *Proc. 11th IEEE Annual Ubiquitous Comput., Electronics and Mobile Comm. Conf.* (IEEE, New York), 0542–0547.

Hurtado S, Ray P, Marculescu R (2019) Bot detection in Reddit political discussion. Ramachandran GS, Ortiz J, eds. *Proc. 4th Internat. Workshop on Social Sensing* (Association for Computing Machinery, New York), 30–35.

Jain G, Sharma M, Agarwal B (2019) Spam detection in social media using convolutional and long short-term memory neural network. *Ann. Math. Artificial Intelligence* 85(1):21–44.

Ji Y, He Y, Jiang X, Cao J, Li Q (2016) Combating the evasion mechanisms of social bots. *Comput. Security* 58:230–249.

Kajino H, Tsuboi Y, Sato I, Kashima H (2012) Learning from crowds and experts. Hoffman J, Selman B, eds. *Proc. Workshops at the 26th AAAI Conf. on Artificial Intelligence* (AAAI, Palo Alto, CA).

Kiene C, Jiang JA, Hill BM (2019) Technological frames and user innovation: Exploring technological change in community moderation teams. *Proc. ACM Human Comput. Interactions*, 1–23.

Kudugunta S, Ferrara E (2018) Deep neural networks for bot detection. *Inform. Sci.* 467:312–322.

Kuo FY, Yin CP (2011) A linguistic analysis of group support systems interactions for uncovering social realities of organizations. *ACM Trans. MIS* 2(1):1–21.

Lapowsky I (2017) It's super hard to find the humans in the FCC's net neutrality comments. *Wired*, https://www.wired.com/story/bots-form-letters-humans-fcc-net-neutrality-comments/.

Lasecki WS, Song YC, Kautz H, Bigham JP (2013) Real-time crowd labeling for deployable activity recognition. Bruckman A, Counts S, Lampe C, Terveen L, eds. *Proc. Conf. Comput. Supported Cooperative Work* (Association for Computing Machinery, New York), 1203–1212.

Leano R, Wang Z, Sarma A (2016) Labeling relevant skills in tasks: Can the crowd help? Blackwell A, Plimmer B, Stapleton G, eds. *Proc. IEEE Sympos. on Visual Languages and Human-Centric Comput.* (IEEE, New York), 185–189.

Levy O, Goldberg Y (2014) Neural word embedding as implicit matrix factorization. Ghahramani Z, Welling M, Cortes C, Lawrence N, Weinberger KQ, eds. *Advances in Neural Information Processing Systems* (MIT Press, Cambridge, MA), 2177–2185.

Liu Y, Liu Z, Chua TS, Sun M (2015) Topical word embeddings. Gunning D, Yeh PZ, eds. *Proc. 29th AAAI Conf. on Artificial Intelligence* (AAAI, Palo Alto, CA).

Liu X, Zhang B, Susarla A, Padman R (2020) Go to YouTube and call me in the morning: Use of social media for chronic conditions. *Management Inform. Systems Quart.* 44(1):257–283.

Ludwig S, de Ruyter K (2016) Decoding social media speak: Developing a speech act theory research agenda. *J. Consumer Marketing* 33(2):124–134.

Lyytinen K (1985) Implications of theories of language for IS. *Management Inform. Systems Quart.* 9(1):61–74.

Maier-Hein L, Ross T, Gröhl J, Glocker B, Bodenstedt S, Stock C, Heim E, et al. (2016) Crowd-algorithm collaboration for large-scale endoscopic image annotation with confidence. Essert C, ed. *Proc. Internat. Conf. on Medical Image Comput. and Comput.-Assisted Intervention* (Springer, Cham, Switzerland), 616–623.

Melis L, Song C, De Cristofaro E, Shmatikov V (2019) Exploiting unintended feature leakage in collaborative learning. Gondree M, Kruegel C, Shacham H, eds. *Proc. IEEE Sympos. on Security and Privacy* (IEEE, New York), 691–706.

Mikolov T, Chen K, Corrado G, Dean J (2013) Efficient estimation of word representations in vector space. IEEE Computer Society, ed. *Proc. Internat. Conf. on Learn. Representations* (IEEE, New York).

Moldovan C, Rus V, Graesser AC (2011) Automated speech act classification for online chat. Visa S, Inoue A, Ralescu A, eds. *MAICS* (Midwest Artificial Intelligence and Cognitive Science, Cincinnati, OH), 23–29.

Nguyen-Dinh LV, Rossi M, Blanke U, Tröster G (2013) Combining crowd-generated media and personal data: Semi-supervised learning for context recognition. Singh VK, Chua T-S, Jain R, eds. *Proc. 1st ACM Internat. Workshop on Personal Data Meets Distributed Multimedia* (Association for Computing Machinery, New York), 35–38.

Oentaryo RJ, Murdopo A, Prasetyo PK, Lim EP (2016) On profiling bots in social media. Spiro E, Ahn Y-Y, eds. *Proc. Internat. Conf. on Social Informatics* (World Academy of Science, Engineering, and Technology), 92–109.

Olney A, Louwerse M, Mathews E, Marineau J, Hite-Mitchell H, Grasser A (2003) Utterance classification in AutoTutor. *Proc. Human Language Tech.*

Pozzana I, Ferrara E (2020) Measuring bot and human behavioral dynamics. *Frontiers Phys.* 8:125.

Qadir A, Riloff E (2011) Classifying sentences as speech acts in message board posts. Barzilay R, Johnson M, eds. *Proc. Conf. on Empirical Methods in Natural Language Processing* (Association for Computational Linguistics, Stroudsburg, PA), 748–758.

Rai A (2017) Editor's comments: Diversity of design science research. *Management Inform. Systems Quart.* 41(1):iii–xviii.

Reimers N, Gurevych I, Reimers N, Gurevych I, Thakur N, Reimers N, Daxenberger J, et al. (2019) Sentence-BERT: Sentence embeddings using Siamese BERT-networks. Padó S, Huang R, eds. *Proc. Conf. on Empirical Methods in Natural Language Processing* (Association for Computational Linguistics, Stroudsburg, PA).

Ruchansky N, Seo S, Liu Y (2017) CSI: A hybrid deep model for fake news detection. Lim E-P, Winslett M, Sanderson M, Fu A, Sun J, Culpepper S, Lo E, Ho J, Donato D, Agrawal R, Zheng Y, Castillo C, Sun A, Tseng VS, Li C, eds. *Proc. ACM Conf. on Inform. and Knowledge Management* (Association for Computing Machinery, New York), 797–806.

Searle JR (1969) *Speech Acts: An Essay in the Philosophy of Language* (Cambridge University Press, Cambridge, UK).

Shi P, Zhang Z, Choo KKR (2019) Detecting malicious social bots based on clickstream sequences. *IEEE Access* 7:28885–28862.

Song K, Tan X, Qin T, Lu J, Liu TY (2019) MASS: Masked sequence to sequence pre-training for language generation. Xing E, Chaudhuri K, Salakhutdinov R, eds. *Proc. Internat. Conf. on Machine Learn.* (Association for Computing Machinery, New York), 5926–5936.

Stieglitz S, Brachten F, Ross B, Jung A-K (2017a) Do social bots dream of electric sheep? A categorisation of social media bot accounts. Jeffery R. ed. *Proc. Australasian Conf. on Inform. Systems* (Australasian Association for Information Systems, Melbourne, Australia), 1–11.

Stieglitz S, Brachten F, Berthel D, Schlaus M, Venetopoulou C, Veutgen D (2017b) Do social bots still act different to humans? Comparing metrics of social bots with those of humans. Jeffery R. ed. *Proc. Internat. Conf. on Social Comput. and Social Media* (Australasian Association for Information Systems, Melbourne, Australia), 379–395.

Subramanian S, Cohn T, Baldwin T (2019) Target based speech act classification in political campaign text. Mihalcea R, Shutova E, Ku L-W, Evang K, Poria S, eds. *Proc. 8th Joint Conf. on Lexical and Computational Semantics* (Association for Computiational Linguistics, Stroudsburg, PA), 273–282.

Subrahmanian V, Azaria A, Durst S, Kagan V, Galstyan A, Lerman K, Zhu L, Ferrara E, Flammini A, Menczer F (2016) The DARPA Twitter bot challenge. *Computer* 49(6):38–46.

Sun C, Huang L, Qiu X (2019) Utilizing BERT for aspect-based sentiment analysis via constructing auxiliary sentence. Burstein J, Doran C, Solorio T, eds. *Proc. NAACL-HLT* (Association for Computiational Linguistics, Stroudsburg, PA), 380–385.

Sundermeyer M, Schlüter R, Ney H (2012) LSTM neural networks for language modeling. Navarro Mesa JL, Ortegoa A, Teixeira A, Perez EH, Morales PQ, Garcia AR, Moreno IG, Toledano DT, eds. *Proc. 13th Annual Conf. of the Internat. Speech Comm. Assoc.* (Elsevier, Amsterdam).

Vaast E, Safadi H, Lapointe L, Negoita B (2017) Social media affordances for connective action: An examination of microblogging use during the Gulf of Mexico oil spill. *Management Inform. Systems Quart.* 41(4):1179–1205.

Varol O, Ferrara E, Davis CA, Menczer F, Flammini A (2017) Online human-bot interactions: Detection, estimation, and characterization. Ruths D, Mason W, Marwick A, Gonzalez-Bailon S, eds. *Proc. 11th AAAI Conf. on Web and Social Media*, (AAAI, Palo Alto, CA).

Wang Y (2017) A new concept using LSTM neural networks for dynamic system identification. Sun J, Rajamani R, eds. *Proc. Amer. Control Conf.* (IEEE, New York), 5324–5329.

Wei F, Nguyen UT (2019) Twitter bot detection using bidirectional long short-term memory neural networks and word embeddings. Joshi J, ed. *Proc. 1st IEEE Internat. Conf. on Trust, Privacy and Security in Intelligent Systems and Appl.* (IEEE, New York), 101–109.

Winograd T, Flores F (1986) *Understanding Computers and Cognition* (Abex, Norwood, NJ).

Wolf T, Chaumond J, Debut L, Sanh V, Delangue C, Moi A, Cistac P, et al. (2020) Transformers: State-of-the-art natural language processing. Wolf T, Debut L, Sanh V, Chaumond J, Delangue C, Moi A, Cistac P, Rault T, Louf R, Funtowicz M, Davison J, Shleifer S, von Platen P, Ma C, Jernite Y, Plu J, Xu C, Le Scao T, Gugger S, Drame M, Lhoest Q, Rush A, eds. *Proc. Conf. on Empirical Methods in Natural Language Processing: System Demonstrations* (Association for Computiational Linguistics, Stroudsburg, PA), 38–45.

Yang KC, Varol O, Davis CA, Ferrara E, Flammini A, Menczer F (2019) Arming the public with artificial intelligence to counter social bots. *Human Behav. Emerging Tech.* 1(1).

Zhu Z, Blanke U, Tröster G (2016) Recognizing composite daily activities from crowd-labelled social media data. *Pervasive Mobile Comput.* 26:103–120.